LESSON PLAN · WEB DEVELOPMENT (GR 6–8) BEGINNER: WEB CREATORS · LEVEL 1

Lesson 2: Lists, links and images

Module 1 | 60 minutes | lists, links, images | Grades 6–11 · ages 11–16

Objective: Add bulleted and numbered lists, links and images with alt text to a web page, and credit the pictures you use.

Success criteria

- I can make bulleted and numbered lists.
- I can add a link with href.
- I can add a picture with helpful alt text.
- I can credit the pictures I use.

Key vocabulary

- **ul**: An unordered (bulleted) list.
- **ol**: An ordered (numbered) list.
- **li**: One item inside a list.
- **link**: Clickable words made with the a tag that take you to another page.
- **alt text**: Words that describe a picture for people who cannot see it.
- **credit**: A line saying who made a picture or where it came from.

You will need

- Projector with Kitabash (lesson video, Web Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: l02-mango-lassi.html, l02-teacher-demo.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: List Relay**	Play List Relay: bullets or numbers? Then the alt-text guessing round.
5 min	**Hook and video**	Show the hook list. Poll: “Bullets or numbers for a recipe’s steps?” Play the video and pause at the image scene to ask who alt text helps.
10 min	**Live demo and build-along**	Run the demo as written (break the image address on purpose), then build the mango lassi page together.
8 min	**Challenge and games**	Web Lab: make a list and a link, then Predict the list, the match game and Order it. Quick competition: fastest correct match wins a shout-out.
17 min	**Project in pairs**	Breakout rooms: pairs build a team or recipe page. Navigator checks every picture has alt text and a credit. Swap roles every 5 minutes.

5 min	**Probe: debug race**	Race to fix “the silent picture”. First learner to explain both bugs wins.
6 min	**Show-and-tell**	Three learners share their page and read their alt text aloud. Class votes on the most helpful alt text with emoji reactions.
4 min	**Wrap-up and exit ticket**	Thumbs check on each success criterion. Exit ticket: picture of a kite with good alt text.

Unplugged warm-up

List Relay (5 min game): call out a topic such as “recipe steps”, “my favourite fruits” or “how to wash hands”. Learners shout “bullets!” (ul) or “numbers!” (ol) depending on whether the order matters. Then describe a mystery picture in one sentence of alt text — others guess what it shows. Best describer wins.

Teaching the idea

Lists come in two kinds: ul for bullets and ol for numbers, and every item sits inside li. The a tag makes a link, and href holds the address. The img tag shows a picture from the address in src; it has no closing tag. Alt text describes the picture for screen-reader users and appears if the image fails. We only use pictures we made or are allowed to share, and we credit them.

Analogy: A link is like a door with an address painted on it: href is the address, and the words are the door handle you click.

Live demo: 1) Type <ul> and , then three <li> lines inside with fruits. Point out the bullets. 2) Change ul to ol in both tags and show the numbers appear. 3) Type Mangoes and explain href is the address and the words between are what you click. Note that links in the preview open in a new tab. 4) Type <img src="https://placehold.co/300×180?text=Mango" alt="A ripe yellow mango"> and point out there is no closing tag. 5) Break the src address on purpose and show the alt text appear where the picture should be. Explain that screen readers read the alt text aloud to blind users. 6) Add a credit line under the picture and talk about only using your own pictures or free ones you can credit.

Questions to ask

Ask	Listen for
When should you use ol instead of ul?	When the order matters, like steps or rankings.
What does href hold?	The address the link takes you to.
Why does img not have a closing tag?	It has no content inside it; it just places a picture.
Who does alt text help?	People using screen readers, and everyone if the picture does not load.

Can you use any picture you find online?	No. Use your own, a placeholder, or a picture that is free to share, and credit it.

Differentiation

Support
Give a printed list template (ul, three li, /ul) to copy. Let learners keep the placeholder image and only change the text= part of its address.

Stretch
Make a list inside a list (a nested ul), add a title="..." tooltip to a link, and add width="200" to an image.

Common misconceptions

Misconception	How to fix it
A list is just lines with dashes typed in	Use ul or ol with li so the browser and screen readers know it is a list.
The img tag needs </img>	img has no closing tag. All its information goes inside the one tag.
Alt text is a title for the picture, like “image1”	Alt text should describe what the picture shows in words.
Any picture on the internet is free to use	Pictures belong to their makers. Use your own or free-to-share ones and credit them.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the HTML for a picture of a kite with good alt text.

Homework
Add a “Top 3” numbered list and one more link to your team or recipe page. Ask someone at home to close their eyes while you read your alt text aloud: could they picture it?

Answer key — Kitabash activities

1. Web Lab: make a list and a link (perform · web lab)

Goals checked: Has a bulleted list · Has at least three list items · Has a link · The link has an address

One solution:

```
<h1>Best monsoon snacks</h1>
<ul>
  <li>Pakoras</li>
  <li>Samosas</li>
  <li>Corn on the cob</li>
  <li>Hot chai</li>
</ul>
<p>Read about <a href="https://en.wikipedia.org/wiki/Pakora">pakoras</a>.</p>
```

2. Predict the list (perform · code output)

3. Match each tag or part to its job (perform · matching)

 → A bulleted list · → A numbered list · → One item in a list · href="..." → The address a link goes to · src="..." → Where a picture lives · alt="..." → Words that describe a picture

4. Order it: add a picture safely (perform · ordering)

Order: Find a picture you made or one that is free to share → Copy the picture's web address → Write an img tag with the address in src → Add alt text that describes the picture → Write a credit line under the picture

5. Project: favourite team or recipe page (perform · web lab)

Goals checked: Has a page title · Has two lists · Has at least five list items · Has a picture with alt text · Has a link · Has enough to read

Example page:

```
<!-- My favourite team page -->
<h1>Go, Mumbai Waves!</h1>
<img src="https://placehold.co/300x180?text=Mumbai+Waves" alt="Placeholder banner with
the words Mumbai Waves">
<p>Picture credit: placeholder image from placehold.co</p>

<h2>Why I love this team</h2>
<ul>
  <li>They never give up, even when they need 30 runs in 2 overs.</li>
  <li>Their spin bowlers are magic.</li>
  <li>The fans sing all match long.</li>
  <li>Blue is my favourite colour.</li>
</ul>

<h2>My top 3 matches</h2>
<ol>
  <li>The final with the last-ball six</li>
  <li>The rain match that went to a super over</li>
  <li>The day the spinners took 8 wickets</li>
</ol>

<h2>Find out more</h2>
<p>Read about the history of <a href="https://en.wikipedia.org/wiki/Cricket">cricket on
Wikipedia</a>.</p>
```

6. Debug it: the silent picture (probe · web lab)

Goals checked: The picture has alt text · Every list item closes

One solution:

```
<h1>My cricket kit</h1>
<img src="https://placehold.co/300x180?text=Cricket+Bat" alt="A wooden cricket bat">
<ul>
  <li>Bat</li>
  <li>Ball</li>
  <li>Pads</li>
</ul>
```

Answer key — worksheet

1. **Key words:** ul = An unordered (bulleted) list. · ol = An ordered (numbered) list. · li = One item inside a list. · link = Clickable words made with the a tag that take you to another page.
2. **Plan your page:** Goals: Has a bulleted list; Has at least three list items; Has a link; The link has an address.
3. **Predict the list:** A. Three lines numbered 1, 2 and 3

4. **Match each tag or part to its job:** → A bulleted list · → A numbered list · → One item in a list · href="..." → The address a link goes to · src="..." → Where a picture lives · alt="..." → Words that describe a picture
5. **Order it: add a picture safely:** 1 Find a picture you made or one that is free to share · 2 Copy the picture's web address · 3 Write an img tag with the address in src · 4 Add alt text that describes the picture · 5 Write a credit line under the picture

Kitabash original curriculum.